

Translations of the Hungarian Physics Olympiad

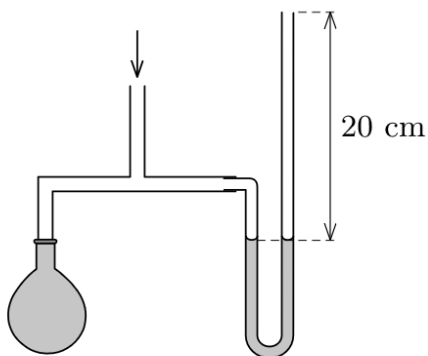
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Problem 1

The elastic behaviour of a balloon can be characterized with the help of the wall's elastic energy. If, to a good approximation, the balloon remains spherical throughout, and its size changes by a factor of λ relative to its unstretched size, then the balloon's elastic energy increases in direct proportion to the expression $2\lambda^2 + \lambda^{-4} - 3$ all the way up to $\lambda \approx 3$ at which value the balloon eventually bursts.

An initially slack balloon is connected to a T-shaped distributor attached to a compressor hose: one outlet is connected to the balloon, while the other outlet is connected to a thin glass tube water manometer, which is fixed as shown in the figure. The water occupies a 20cm long section of the tube. After starting the compressor, we observe that the liquid levels appear to remain at their original positions, while the balloon's diameter increases by 5%. What will happen after this, once the liquid levels are able to shift slowly by 5cm to accommodate their new equilibrium?



Problem 2

An aluminium tube of radius R , length $L \gg R$ and wall thickness $d \ll R$ is placed in a state of weightlessness, in a homogenous magnetic field of induction B perpendicular to its axis of symmetry. The tube is set spinning about its own axis with angular velocity ω_0 and then released.

- Sketch, on a drawing of the tube's unrolled (flattened) surface, the induced current pattern that forms in the tube.
- Describe the tube's motion as a function of time.

Problem 3

In a tube, two converging lenses with focal lengths $f_1 = 50\text{cm}$ and $f_2 = 10\text{cm}$ are placed symmetrically, as shown in the left-hand figure. We managed to set up the lenses so that, looking through the optical system of objects, we see exactly what we would see through an empty tube.

a) What is the distance d and s between the lenses?

b) If we build this optical system without the tube, and place it between our eye and the appropriate location, so that as shown in the right-hand figure, we can make part of our hand "disappear" from view - explain this phenomenon!

